

Project 4: Preference Elicitation

Human-Centric Artificial Intelligence

Objective

The objective of this project is to design a rigorous user study comparing two methods for eliciting movie preferences: repeated pairwise choices and ranking sets of ten movies. The project also implements the complete participant-facing Django interface that could be used to conduct the proposed study.

***Important:** The user study is designed as part of this project but is **not conducted**. Therefore, this report presents the experimental protocol and analysis plan but does not report empirical participant results.*

1. Introduction

The purpose of preference elicitation is to infer what a user likes from a relatively small number of interactions. In this project, the application domain is movie recommendation. For a movie represented by a feature vector x , the user's utility is modeled as:

$$U(x) = w^T x$$

The vector w represents the user's latent movie preferences. Positive components indicate characteristics that increase predicted utility, while negative components indicate characteristics that decrease predicted utility. The preference elicitation process therefore aims to estimate w from limited user feedback.

2. Dataset

The project uses the IMDB 5000 Movie Dataset. The dataset contains metadata for approximately 5000 movies. Unlike traditional collaborative-filtering datasets, it does not contain ratings from individual users. Consequently, the system cannot directly learn a user's preferences from historical ratings. Instead, preference information must be obtained through interactions with the participant.

The implemented system uses movie metadata both for the numerical preference representation and for presenting understandable information to participants in the interface.

3. Task 1 - Movie Feature Representation

3.1 Selected Features

Each movie is represented using genre information, duration, release year, IMDb score and content rating. The objective is to capture several meaningful dimensions of movie preference while keeping the feature space sufficiently compact to estimate a user model from a limited number of interactions.

Feature	Representation	Justification
Genres	Multi-hot encoding	A movie can belong to several genres and users frequently have genre-specific preferences.
Duration	Standardized numerical	Captures preferences for shorter or longer movies.
Release year	Standardized numerical	Captures preference for older or newer movies.
IMDb score	Standardized numerical	Provides information about the general reception of a movie.
Content rating	One-hot encoding	Represents categorical information about the intended audience/content category.

Director and actor identities are intentionally not included in the preference vector. These attributes have a very large number of distinct values and would therefore create a high-dimensional sparse representation. Because the purpose of the project is to estimate the latent preference vector from relatively few interactions, such a representation would make parameter estimation considerably more difficult. Director information may still be displayed to the participant without being included in the learned vector.

3.2 Feature Extraction Method

The feature extraction procedure implemented in `feature_extraction.py` performs the following preprocessing steps:

Step	Operation
1	Load the IMDB CSV file with Pandas.
2	Remove observations without movie titles and remove duplicate movie records.
3	Split the genre string using the separator and create one binary indicator for every genre. Since several indicators may equal one simultaneously, this is multi-hot encoding.
4	Convert duration, release year and IMDb score to numeric values. Missing values are replaced by the median of the corresponding feature.
5	Standardize the numerical features using $z = (x - \text{mean}) / \text{standard deviation}$.
6	Replace missing content ratings with an Unknown category and one-hot encode the resulting categorical variable.
7	Concatenate genre, numerical and content-rating features into the final numerical feature matrix X.

For movie i , the resulting row x_i of this feature matrix is used directly in the utility function $U(x_i) = w^T x_i$.

4. Task 2 - Preference Model

4.1 Bradley-Terry Model for Pairwise Choices

In the pairwise interface, the participant is shown two movies i and j and chooses the movie they would rather watch. Their utilities are $U_i = w^T x_i$ and $U_j = w^T x_j$.

$$P(i > j \mid w) = \exp(w^T x_i) / [\exp(w^T x_i) + \exp(w^T x_j)]$$

Equivalently, the probability may be written using the logistic function as:

$$P(i > j \mid w) = \text{sigmoid}(w^T (x_i - x_j))$$

The parameter vector w is estimated by maximizing the likelihood of the participant's observed choices. In the implementation this is formulated as minimization of the negative log-likelihood with L2 regularization.

4.2 Extension to Rankings: Plackett-Luce Model

The second interface does not provide a single pairwise comparison. Instead, the participant provides a complete ranking:

$$i_1 > i_2 > \dots > i_n$$

To model this information, the Bradley-Terry idea is extended using the Plackett-Luce model:

$$P(i_1 > i_2 > \dots > i_n \mid w) = \text{Product}_{k=1}^{n-1} \left\{ \frac{\exp(w^T x_{i_k})}{\sum_{j=k}^n \exp(w^T x_{i_j})} \right\}$$

The model interprets a ranking as a sequence of choices. First, i_1 is selected as the most preferred movie from all n alternatives. Next, i_2 is selected from the remaining $n-1$ alternatives. This continues until the ranking has been constructed.

This formulation is particularly appropriate because when only two movies are present, it reduces to the Bradley-Terry pairwise probability. Consequently, both elicitation interfaces can use the same movie representation, the same linear utility function and the same interpretation of the latent preference vector w .

4.3 Parameter Estimation

Both models are estimated by maximum likelihood. The implementation minimizes the negative log-likelihood using numerical optimization and adds an L2 penalty to reduce overfitting when relatively few preference observations are available.

5. Task 3 - User Study Design

5.1 Research Question

Research question: Which preference elicitation method learns a new user's movie preferences more effectively and efficiently: repeated pairwise choices or rankings of ten movies?

5.2 Hypotheses

Primary null hypothesis H0: There is no difference between the pairwise and ranking elicitation methods in held-out preference prediction performance.

Primary alternative hypothesis H1: The ranking-based method produces lower held-out negative log loss than the pairwise method when both methods use the same movie exposure budget.

Secondary hypothesis H2: Ranking ten movies simultaneously is perceived as more mentally demanding than making pairwise movie choices.

5.3 Experimental Design

A **within-subject design** is proposed. Every participant uses both preference elicitation interfaces. This design is appropriate because movie tastes differ considerably between individuals; evaluating both methods on the same participant reduces variability due to between-person differences.

A possible disadvantage of a within-subject study is the presence of learning, fatigue and order effects. To reduce these effects, the condition order is counterbalanced. Participants are randomly assigned with approximately equal probability to one of the following sequences:

Group	Condition order
A	Pairwise -> Ranking
B	Ranking -> Pairwise

5.4 Independent and Dependent Variables

Type	Variable
Independent variable	Preference elicitation method: Pairwise vs. Ranking.
Primary dependent variable	Held-out negative log loss of predicted movie preferences.
Secondary dependent variable	Held-out pairwise prediction accuracy.
Secondary dependent variable	Time required to complete each elicitation condition.
Subjective measure	Ease of expressing preferences.
Subjective measure	Perceived mental demand.
Subjective measure	Confidence that the responses represent the participant's actual preferences.

5.5 Interaction Budget

The two elicitation conditions are assigned an equal movie exposure budget. The pairwise condition contains 15 comparisons, exposing the participant to 30 movies. The ranking condition contains three ranking tasks with 10 movies each, also exposing the participant to 30 movies.

Stage	Number of tasks	Movie exposure
Pairwise elicitation	15 pairwise choices	30 movies

Stage	Number of tasks	Movie exposure
Ranking elicitation	3 rankings x 10 movies	30 movies
Held-out evaluation	10 pairwise choices	20 new movies

5.6 Movie Selection

Movies are selected uniformly at random from the available dataset. For a participant, the implementation samples the movies used in the elicitation and evaluation stages without replacement. The 20 movies used in the ten held-out evaluation pairs are therefore not used in either training condition.

This separation is important because both learned models should be evaluated on preferences that were not directly observed during preference elicitation.

A possible future extension would replace uniform random sampling with an adaptive strategy that selects movies expected to provide more information about the user's preference vector.

5.7 Participants and Recruitment

If the study were conducted, a practical target would be approximately 30-40 adult participants. This is proposed as a feasible sample for a course-scale within-subject study; a formal power analysis should be performed before a real scientific deployment if a specific minimum detectable effect is required.

Participants could be recruited through university mailing lists, student networks and online announcements. Participants should be at least 18 years old and sufficiently familiar with movies to express meaningful preferences. Machine-learning expertise is not required.

5.8 Complete Study Procedure

Step	Procedure
1	Present participant information describing the purpose and procedure of the study.
2	Obtain informed consent.
3	Generate an anonymous participant identifier.
4	Randomly assign the participant to Pairwise -> Ranking or Ranking -> Pairwise.
5	Run the first elicitation condition and record responses and completion time.
6	Administer a short post-condition questionnaire.
7	Run the second elicitation condition and record responses and completion time.
8	Administer the same questionnaire after the second condition.
9	Present ten new held-out movie pairs and record which movie is preferred in each pair.
10	Fit one preference vector using the pairwise responses and another using the ranking responses.
11	Use both models to predict the same held-out choices.
12	Compute objective performance metrics and compare the two conditions together with subjective questionnaire scores and completion times.
13	Display the completion/debriefing screen.

5.9 Evaluation Metrics

Held-out accuracy: the proportion of the ten unseen pairwise preferences that the learned model predicts correctly.

Held-out negative log loss: evaluates not only whether the prediction is correct but also whether the model assigns high probability to the participant's actual choice. This is used as the primary predictive metric.

Completion time: measures the time required to complete each elicitation method.

Subjective evaluation: participants rate ease of preference expression, mental demand and confidence using five-point response scales.

5.10 Statistical Analysis Plan

Because every participant completes both interfaces, measurements from the two conditions are paired. For the primary analysis, held-out log loss from the ranking model is compared with held-out log loss from the pairwise model.

If the paired differences are approximately normally distributed, a paired t-test can be used. If this assumption is not appropriate, the Wilcoxon signed-rank test provides a non-parametric alternative.

The same paired-analysis logic can be used for completion time and questionnaire measures. Descriptive statistics should also be reported for all measures.

5.11 Ethical and Data-Protection Considerations

Participants should receive sufficient information about the study before taking part and should explicitly provide informed consent. Participation should be voluntary and participants should be informed that they may stop the study at any time without penalty.

The implemented prototype does not request a participant's name or email address. Instead, a random identifier is stored with the interaction data. Collected information should be restricted to what is required for the study, including preference responses, completion times, condition order and questionnaire responses.

If the experiment were conducted in practice, participants should also be informed about the purpose of data collection, storage duration, access rights and the procedure for requesting deletion or withdrawal of their data.

6. Task 4 - Interactive User Study Interface

The complete participant-facing study is implemented as a Django application. The interface is designed so that the experimental protocol described above could be run with real participants, although conducting the study itself is outside the scope of this project.

Page	Purpose
Landing page	Provides access to this PDF report and the Start Study button.
Study introduction	Displays participant information and obtains consent.
Pairwise interface	Displays two movies and asks which movie the participant would rather watch.
Ranking interface	Displays ten movies and allows the participant to reorder them from most preferred to least preferred.
Questionnaire	Records subjective evaluation after each elicitation condition.
Evaluation interface	Collects ten held-out pairwise choices using movies not used during elicitation.
Completion page	Marks the end of the study and can display model evaluation information for demonstration purposes.

The application stores study state using an anonymous StudySession object. Pairwise responses, ranking responses, held-out choices, condition order, questionnaire responses and model estimates can therefore be associated with the same anonymous participant session.

7. Conclusion

This project proposes and implements a complete protocol for comparing two preference elicitation strategies for movie recommendation. A compact metadata-based movie representation is shared by both methods. Pairwise preferences are modeled using Bradley-Terry, while complete rankings are modeled using the Plackett-Luce extension.

The proposed within-subject experiment compares both methods under an equal movie exposure budget and evaluates them on the same unseen preference choices. Counterbalancing, objective prediction metrics, completion time and subjective measures are included to provide a more complete comparison of effectiveness and user effort.

No real participants are required for the project submission; the implemented Django interface demonstrates how the proposed study could be conducted in practice.